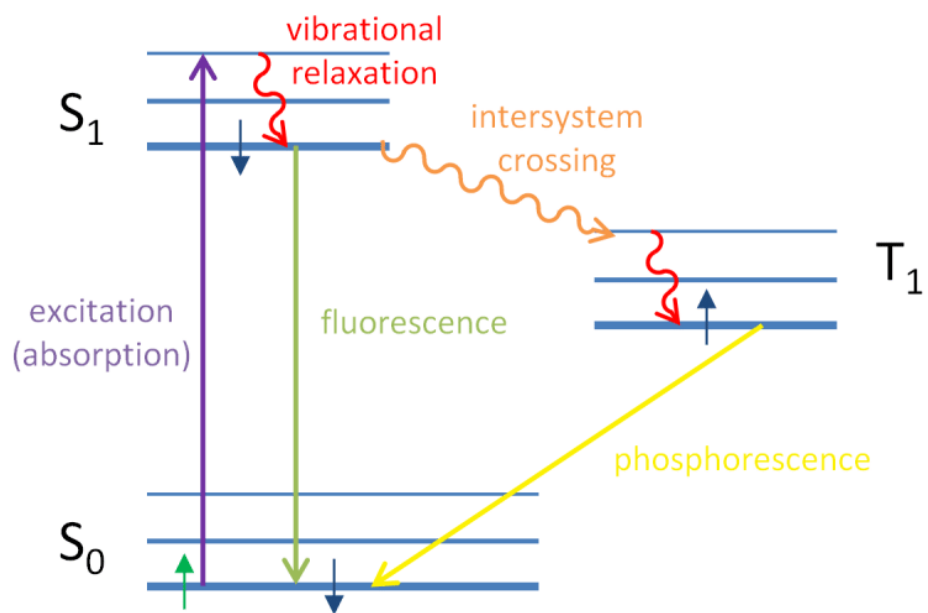


Photoactive molecules and Applications



Singlet and triplet excited states



*Jablonski Diagram
(simplified version)*



Chromophores

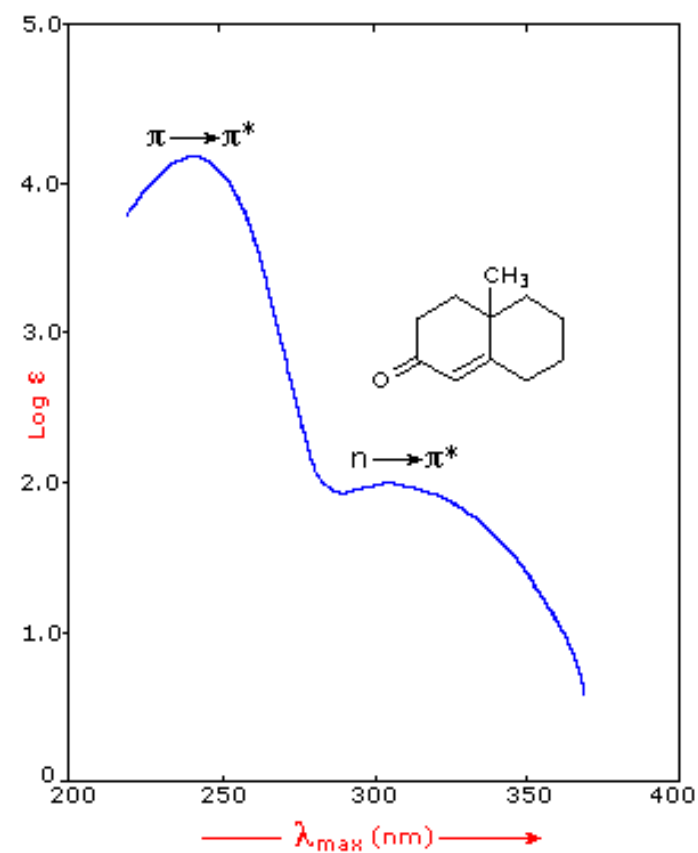
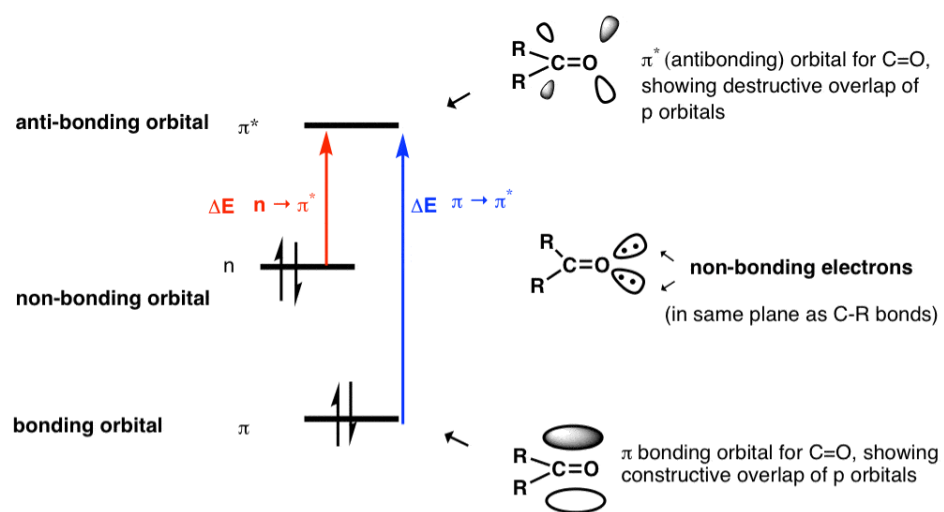
- Specific parts of molecule that can absorb light
- Example: spices such as turmeric, red chilli

Absorbs light / Emits complementary light

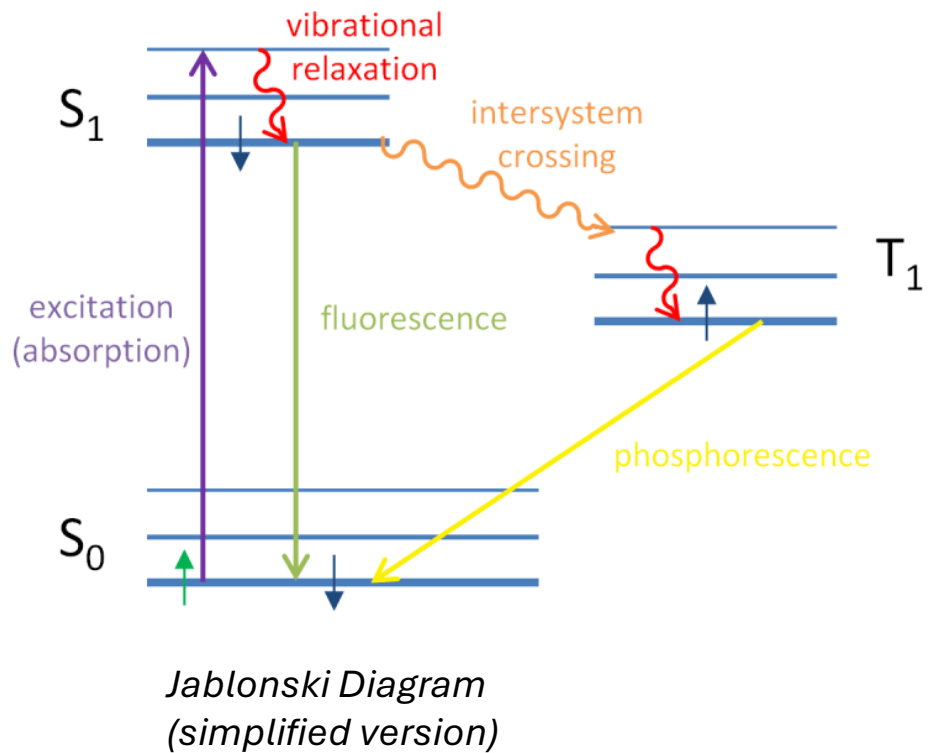


UV-visible spectra

Simplified Molecular Orbital Diagram for Acetone (Propanone)



Singlet and triplet excited states

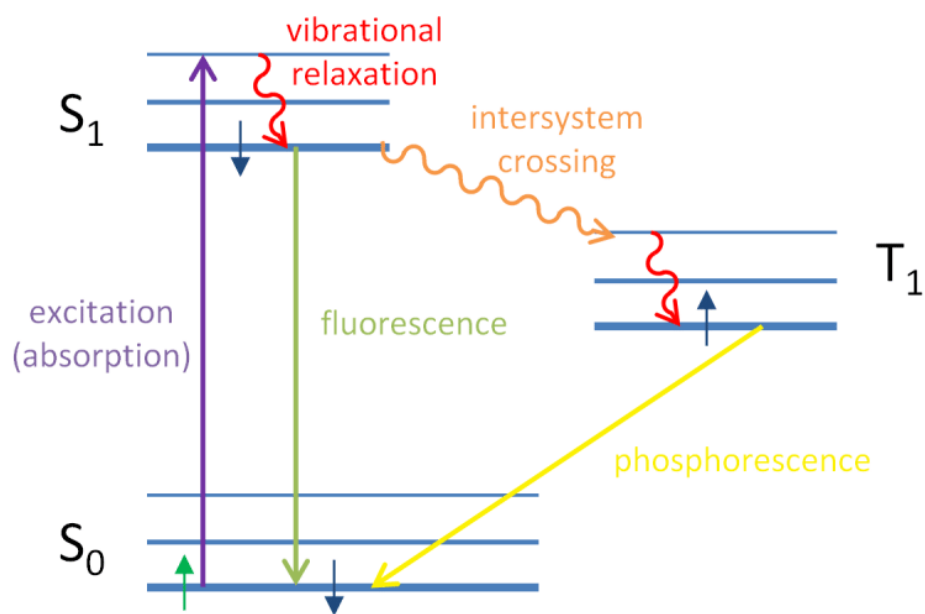


Fluorophores



Refer previous slides

Singlet and triplet excited states



*Jablonski Diagram
(simplified version)*

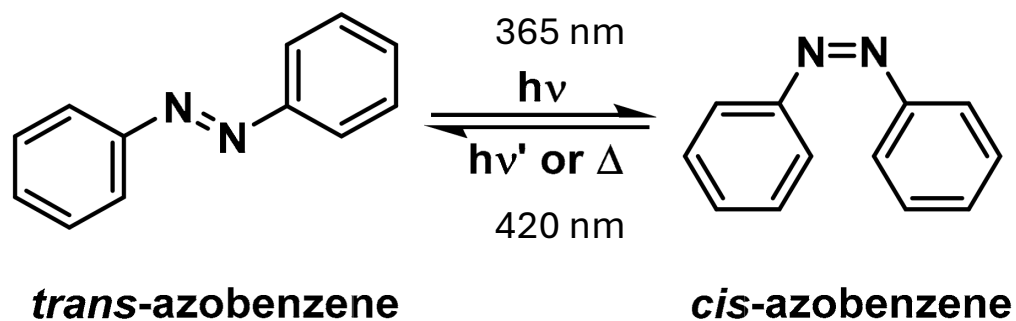
From triplet state – molecules can react

Bond breaking

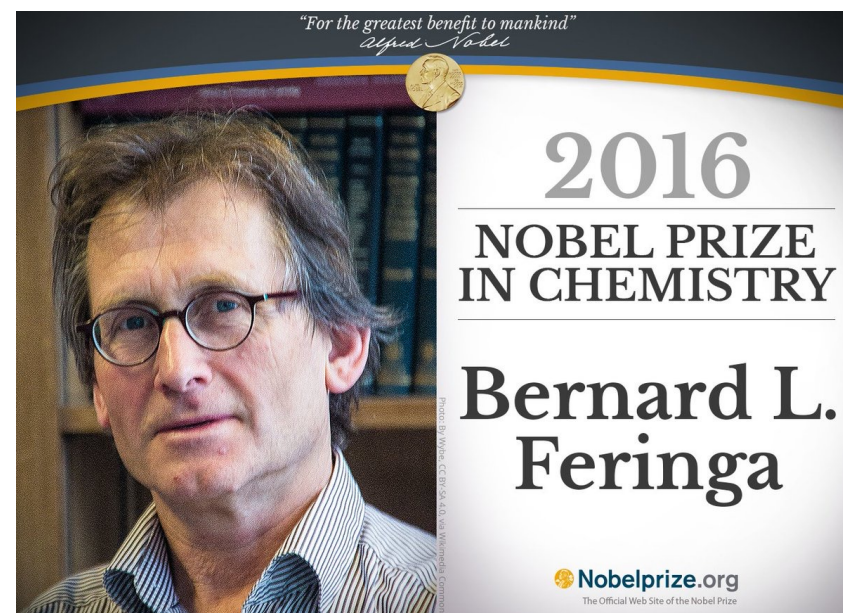
Bond making

With the help of light

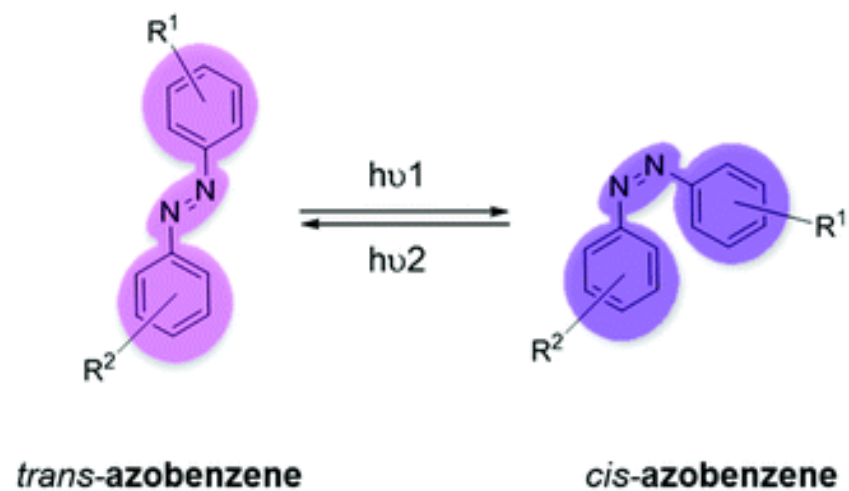
Cis to trans isomerization



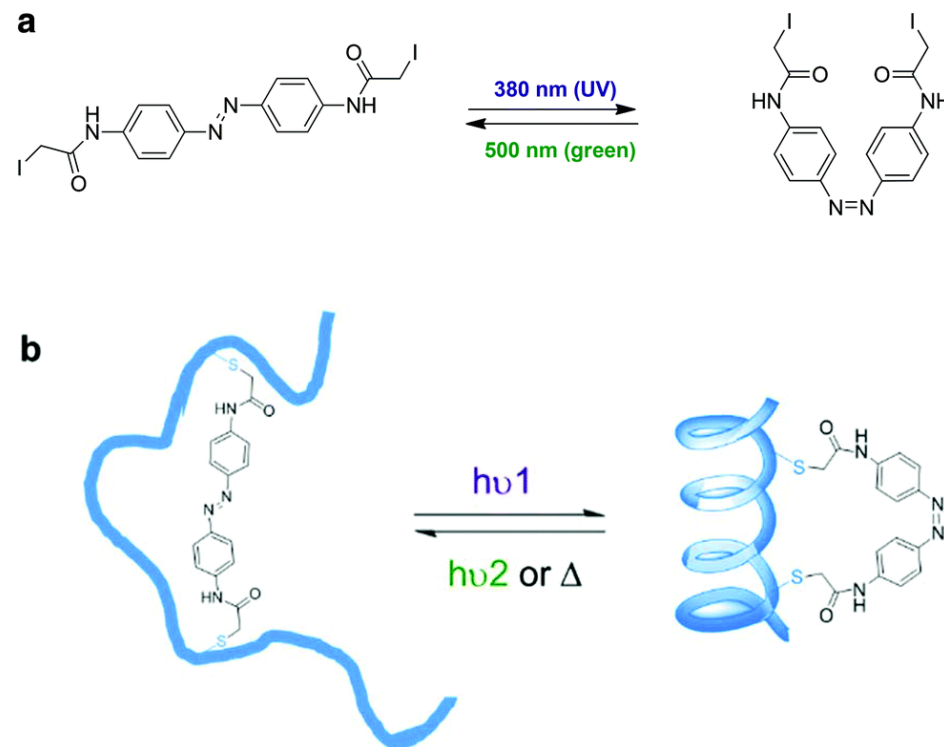
Photoswitches



Azobenzene Photoswitches



Change in shape

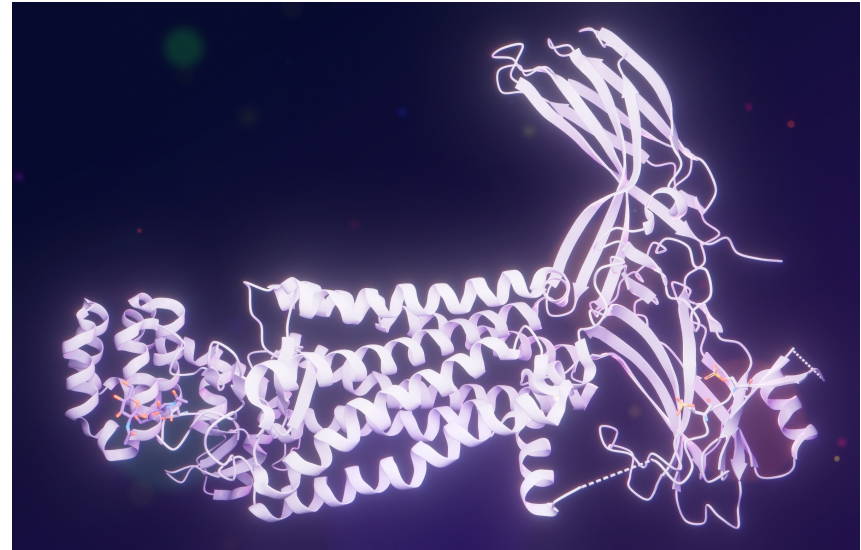




Our retina contains light-sensitive molecules

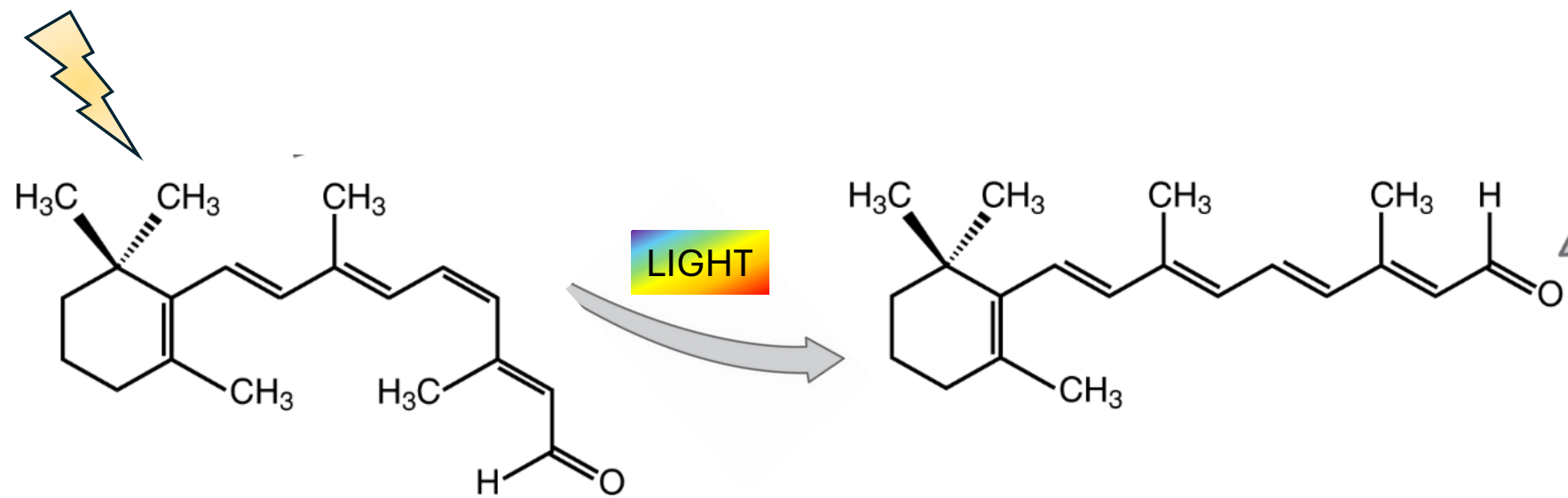
<https://www.youtube.com/watch?v=EgapLhkUP-c>

Rhodopsin



A light-sensitive G-protein-coupled receptor (GPCR) pigment located in the rod cells of the retina

Protein (opsin) + chromophore (retinal)



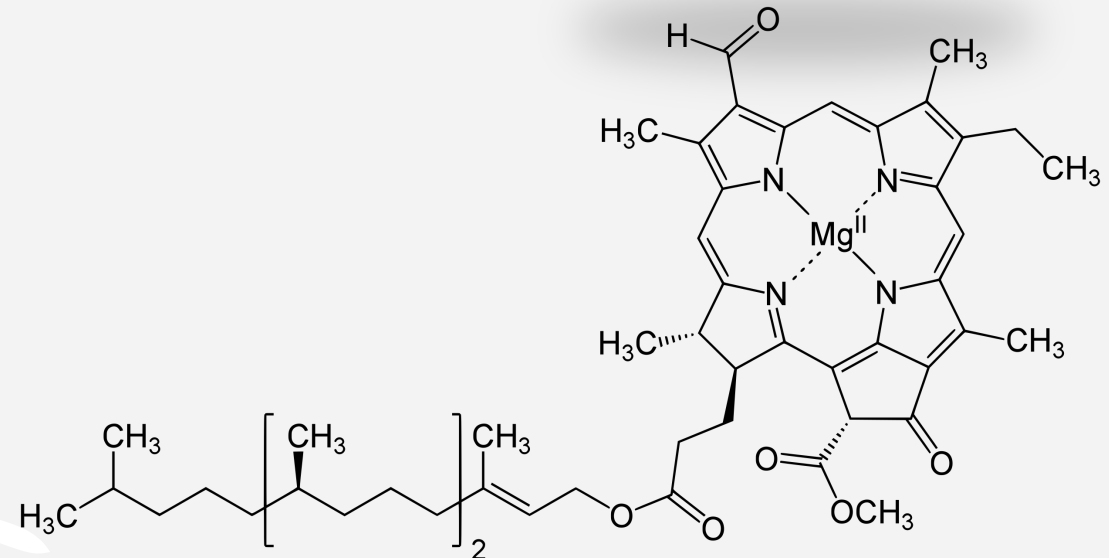
(a) 11-*cis*-retinal

(b) all-*trans*-retinal

*Change in cis-trans configuration triggers change in protein (opsin) conformation
This change in protein conformation triggers signal*

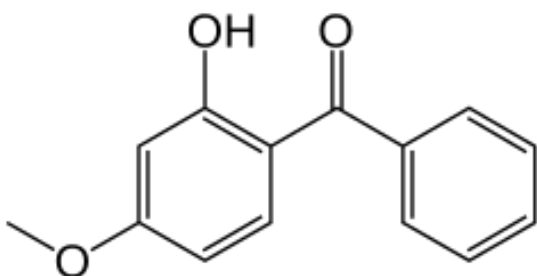
Chlorophyll: Chlorin Magnesium Complex

- The function of the reaction center of chlorophyll is to absorb light energy and transfer it to other parts of the photosystem
- It absorbs light strongly in the blue portion as well as in the red portion of the electromagnetic spectrum
- Donates high-energy electrons to substrates – electron transport chain – photosystems
- Essential for photosynthesis in plants and bacteria

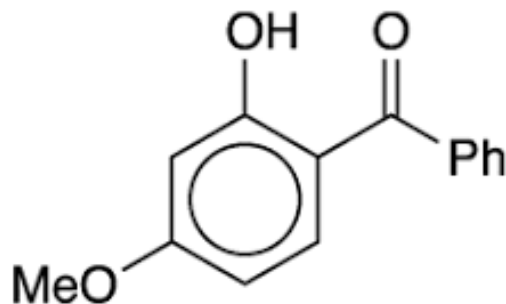


Sunscreen/ UV filters

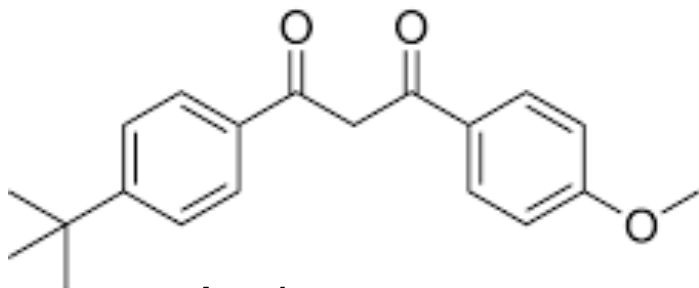
Organic molecules that absorb UV radiation



Oxybenzone



Octinoxate



Avobenzone

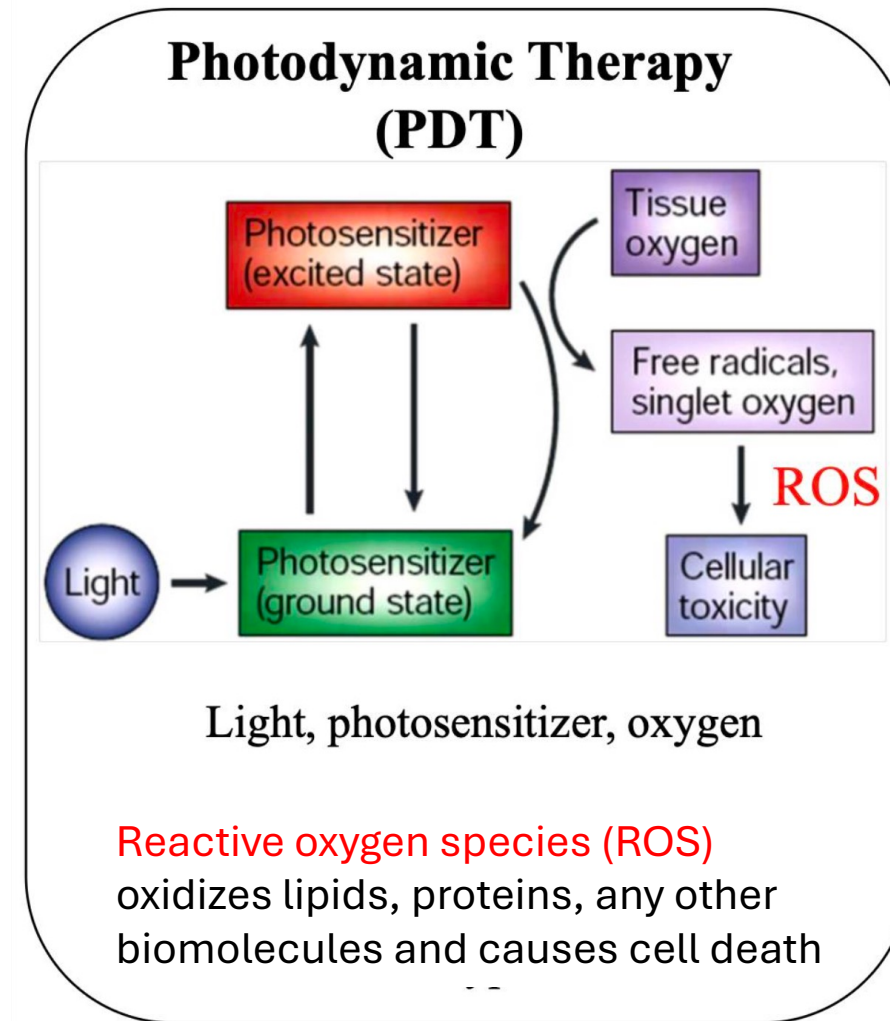
Zinc oxide, titanium dioxide



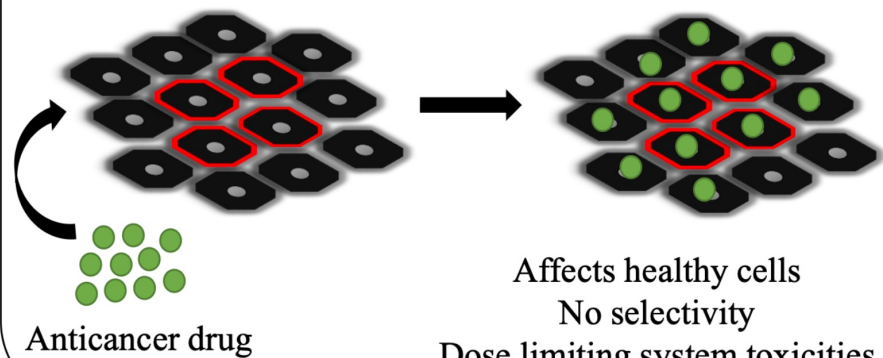
Complete bond
breakage on light
activation

Kill cancer cells with
light activation

Hydroxyl radical,
Superoxide radicals
Peroxide radicals
Hydrogen peroxide

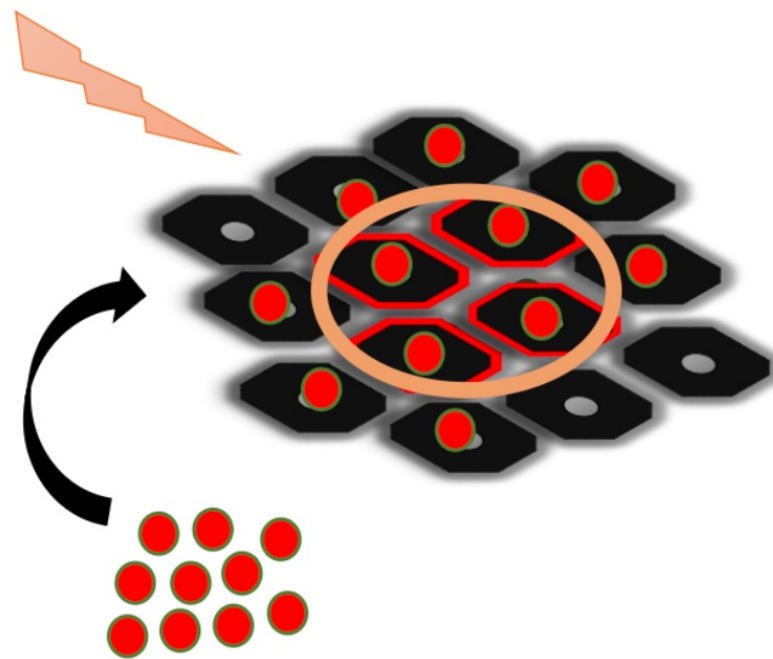


Chemotherapy



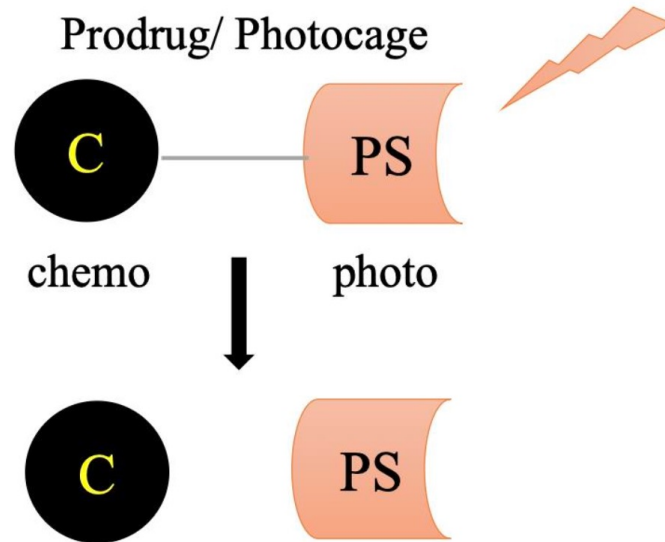
Affects healthy cells
No selectivity
Dose limiting system toxicities

Phototherapy



Molecules are only
activated by light

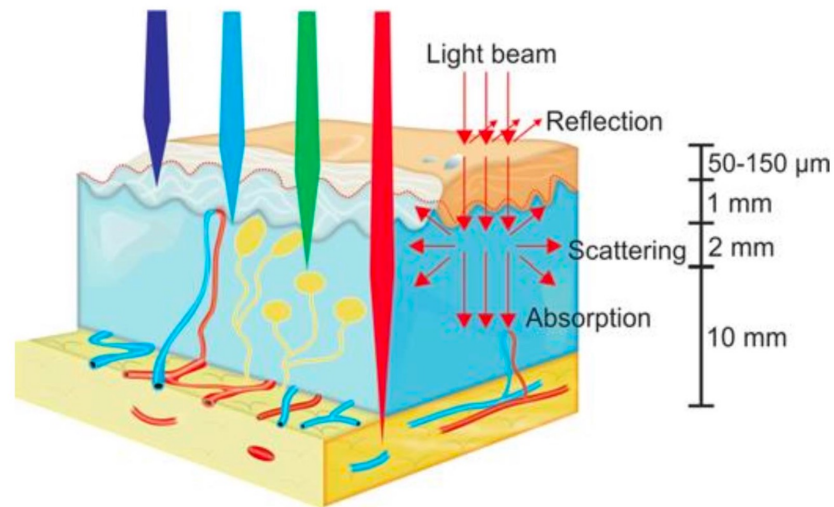
Photoactivated Chemotherapy (PACT)/ Photouncaging



- Oxygen independent mechanism effective for core tumors
 - Use of photosensitizer for Combination Therapy

Far Red and Near Infrared Absorptions

Penetration of light of different wavelength through tissue



- Far red and NIR I (~650-850 nm) activation is desired for phototherapies
- Red light is biocompatible and has superior tissue penetration
- More relevant for clinical applications

BULK ELEMENTS

	BULK NONMETALS
	BULK METALS

TRACE ELEMENTS

	TRACE NONMETALS
	TRACE METALS
	ULTRA-TRACE METALS
	TRACE ELEMENTS FOR SOME SPECIES

1 H Hydrogen																	2 He Helium	
3 Li Lithium	4 Be Beryllium																	10 Ne Neon
11 Na Sodium	12 Mg Magnesium																	18 Ar Argon
19 K Potassium	20 Ca Calcium	21 Sc Scandium	22 Ti Titanium	23 V Vanadium	24 Cr Chromium	25 Mn Manganese	26 Fe Iron	27 Co Cobalt	28 Ni Nickel	29 Cu Copper	30 Zn Zinc	31 Ga Gallium	32 Ge Germanium	33 As Arsenic	34 Se Selenium	35 Br Bromine	36 Kr Krypton	
37 Rb Rubidium	38 Sr Strontium	39 Y Yttrium	40 Zr Zirconium	41 Nb Niobium	42 Mo Molybdenum	43 Tc Technetium	44 Ru Ruthenium	45 Rh Rhodium	46 Pd Palladium	47 Ag Silver	48 Cd Cadmium	49 In Indium	50 Sn Tin	51 Sb Antimony	52 Te Tellurium	53 I Iodine	54 Xe Xenon	
55 Cs Cesium	56 Ba Barium	57 La Lanthanum	72 Hf Hafnium	73 Ta Tantalum	74 W Tungsten	75 Re Rhenium	76 Os Osmium	77 Ir Iridium	78 Pt Platinum	79 Au Gold	80 Hg Mercury	81 Tl Thallium	82 Pb Lead	83 Bi Bismuth	84 Po Polonium	85 At Astatine	86 Rn Radon	
87 Fr Francium	88 Ra Radium	89 Ac Actinium	104 Rf Rutherfordium	105 Db Dubnium	106 Sg Seaborgium	107 Bh Bohrium	108 Hs Hassium	109 Mt Meitnerium	110 Ds Darmstadtium	111 Rg Roentgenium	112 Cn Copernicium	113 Nh Nihonium	114 Fl Flerovium	115 Mc Moscovium	116 Lv Livermorium	117 Ts Tennessine	118 Og Oganesson	

BULK METALS

TRACE METALS

ULTRA-TRACE METALS

TRACE ELEMENTS FOR SOME SPECIES

5
B
Boron

6
C
Carbon

7
N
Nitrogen

8
O
Oxygen

9
F
Fluorine

10
Ne
Neon

13
Al
Aluminium

14
Si
Silicon

15
P
Phosphorus

16
S
Sulfur

17
Cl
Chlorine

18
Ar
Argon

31
Ga
Gallium

32
Ge
Germanium

33
As
Arsenic

34
Se
Selenium

35
Br
Bromine

36
Kr
Krypton

49
In
Indium

50
Sn
Tin

51
Sb
Antimony

52
Te
Tellurium

53
I
Iodine

54
Xe
Xenon

81
Tl
Thallium

82
Pb
Lead

83
Bi
Bismuth

84
Po
Polonium

85
At
Astatine

86
Rn
Radon

113
Nh
Nihonium

114
Fl
Flerovium

115
Mc
Moscovium

116
Lv
Livermorium

117
Ts
Tennessine

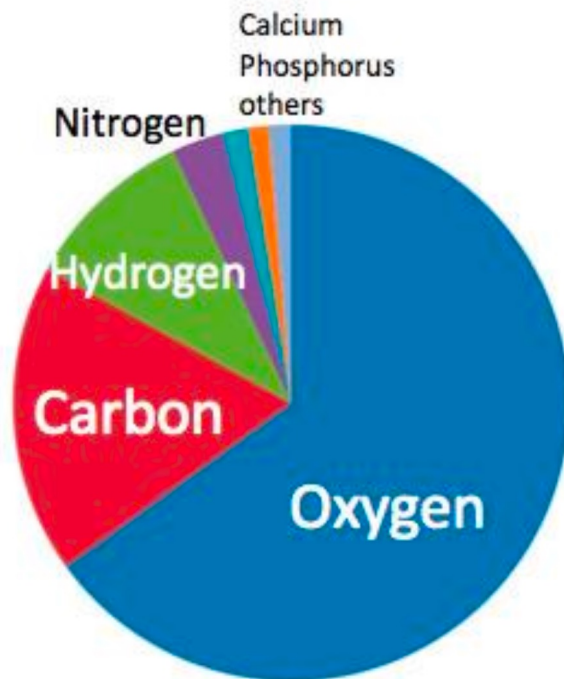
118
Og
Oganesson

Bioinorganic
Chemistry

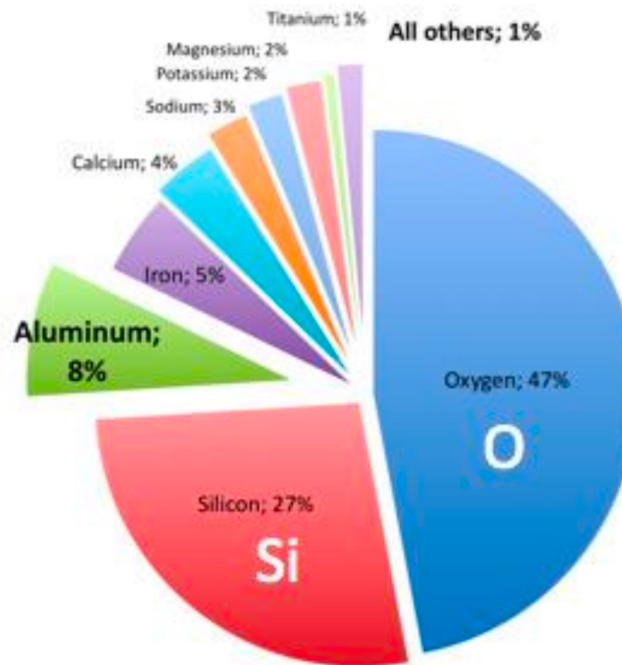
Bioinorganic
Chemistry

Abundance of the Elements:

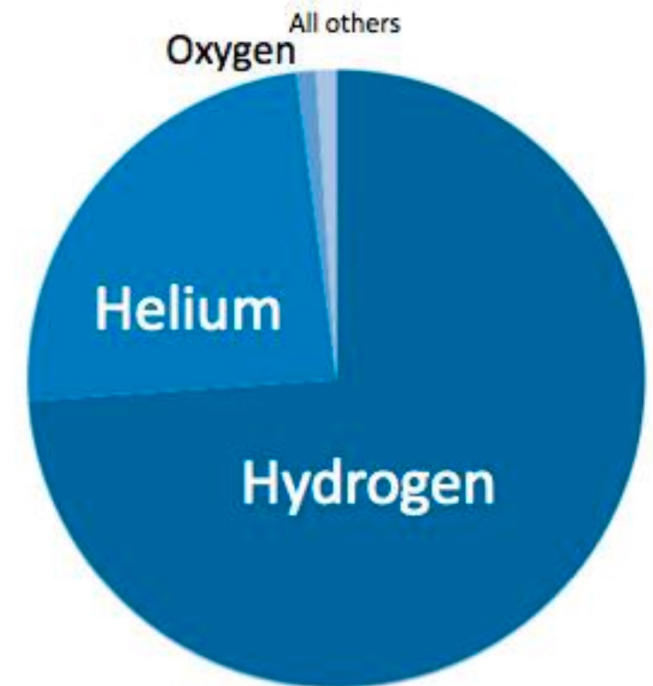
In Life



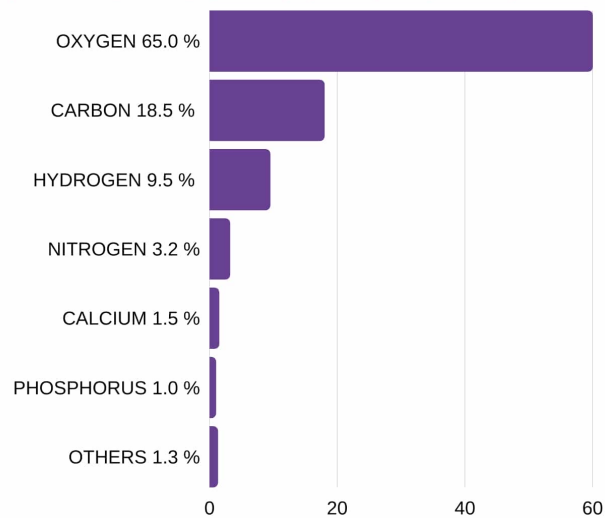
On Earth
(Earth's crust)



In the Universe



CHEMICAL COMPOSITION OF HUMAN BODY



Molecular Formula Human

$\text{H}375,000,000\text{O}132,000,000\text{C}85,700,000\text{N}6,430,000\text{Ca}1,500,000\text{P}1,020,000\text{S}206,000\text{Na}183,000\text{K}177,000\text{Cl}127,000\text{Mg}40,000\text{Si}38,600\text{Fe}2,680\text{Zn}2,110\text{Cu}76\text{I}14\text{Mn}13\text{F}13\text{Cr}7\text{Se}4\text{Mo}3\text{Co}1$

(Stern-Elser) [empirical] april 2000

BULK ELEMENTS



TRACE ELEMENTS

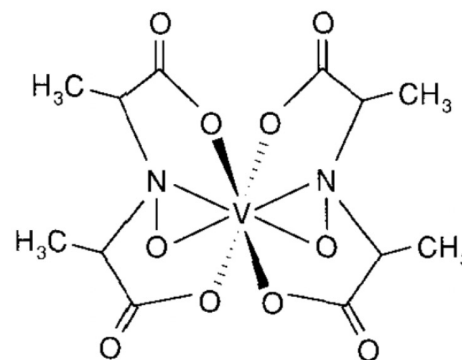


1 H Hydrogen																	2 He Helium
3 Li Lithium	4 Be Beryllium															10 Ne Neon	
11 Na Sodium	12 Mg Magnesium															18 Ar Argon	
19 K Potassium	20 Ca Calcium	21 Sc Scandium	22 Ti Titanium	23 V Vanadium	24 Cr Chromium	25 Mn Manganese	26 Fe Iron	27 Co Cobalt	28 Ni Nickel	29 Cu Copper	30 Zn Zinc	31 Ga Gallium	32 Ge Germanium	33 As Arsenic	34 Se Selenium	35 Br Bromine	36 Kr Krypton
37 Rb Rubidium	38 Sr Strontium	39 Y Yttrium	40 Zr Zirconium	41 Nb Niobium	42 Mo Molybdenum	43 Tc Technetium	44 Ru Ruthenium	45 Rh Rhodium	46 Pd Palladium	47 Ag Silver	48 Cd Cadmium	49 In Indium	50 Sn Tin	51 Sb Antimony	52 Te Tellurium	53 I Iodine	54 Xe Xenon
55 Cs Cesium	56 Ba Barium	57 La Lanthanum	72 Hf Hafnium	73 Ta Tantalum	74 W Tungsten	75 Re Rhenium	76 Os Osmium	77 Ir Iridium	78 Pt Platinum	79 Au Gold	80 Hg Mercury	81 Tl Thallium	82 Pb Lead	83 Bi Bismuth	84 Po Polonium	85 At Astatine	86 Rn Radon
87 Fr Francium	88 Ra Radium	89 Ac Actinium	104 Rf Rutherfordium	105 Db Dubnium	106 Sg Seaborgium	107 Bh Bohrium	108 Hs Hassium	109 Mt Meitnerium	110 Ds Darmstadtium	111 Rg Roentgenium	112 Cn Copernicium	113 Nh Nihonium	114 Fl Flerovium	115 Mc Moscovium	116 Lv Livermorium	117 Ts Tennessine	118 Og Oganesson

Vanadium

Often utilized as vanadate (like phosphates)
Found in plants
Also present in certain organisms
Redox state varies largely

Ascidians: *Ascidia nigra*: Tunichrome
Mushroom *Amanita muscaria* (Amavadinine)



Oxidation state of vanadium?

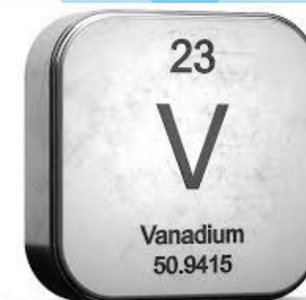
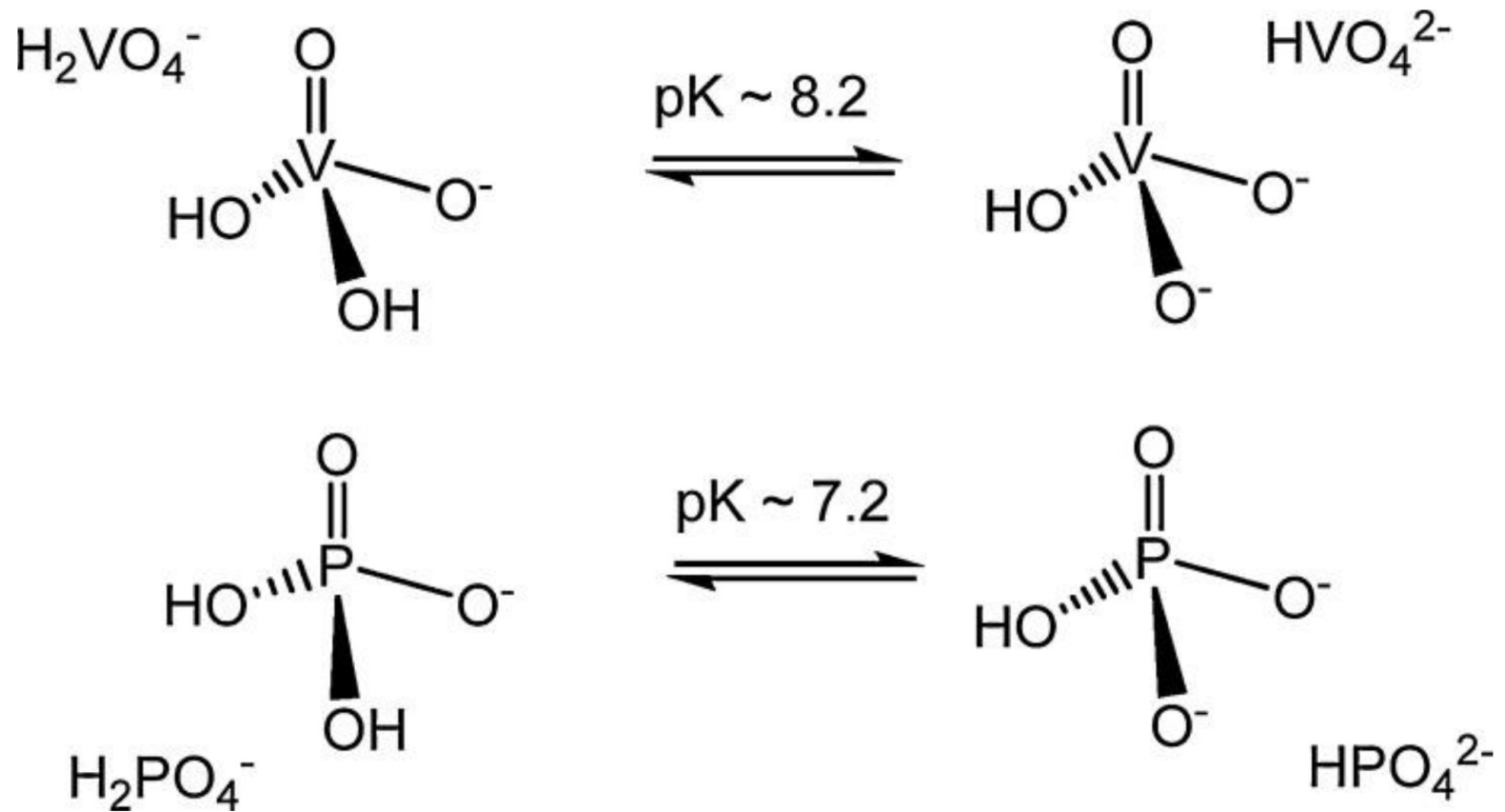


Figure 1.7
A structure proposed for amavadinine.

Like phosphates, vanadates also exist in our body



Chromium

+3 to +6 oxidation states

Cr(III) is relatively biocompatible than Cr(VI). The higher oxidation state is lethal

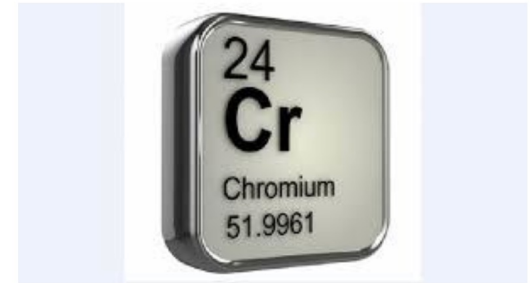
The most mysterious ultra trace element

Present in Glucose tolerance factor

Glucose tolerance improved when supplemented with chromium chloride (150-250 mg of Cr)

Inorganic salts of chromium are relatively unavailable for cells

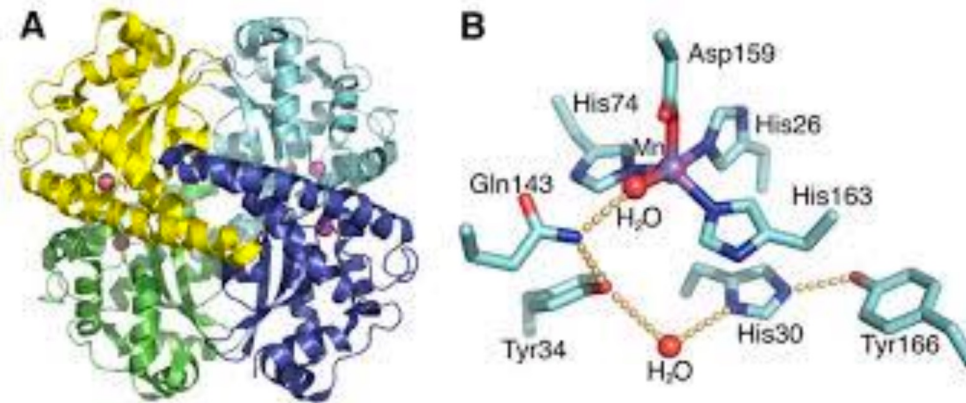
Brewer's yeast can convert the chromium into usable forms – used in diet supplementation for GT



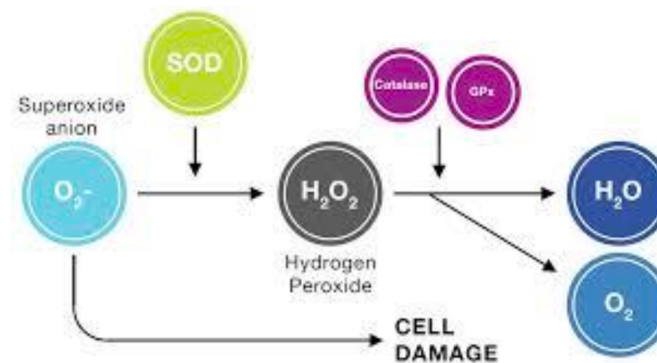
Manganese



Mn - SOD

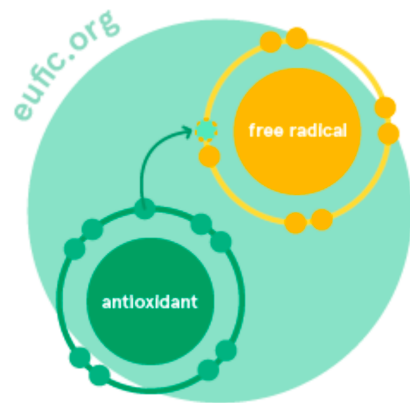


Superoxide dismutase



The redox homeostasis of the cell

functions of manganese



protects our cells
against damage from
free radicals



helps form & keep strong
bones & cartilage



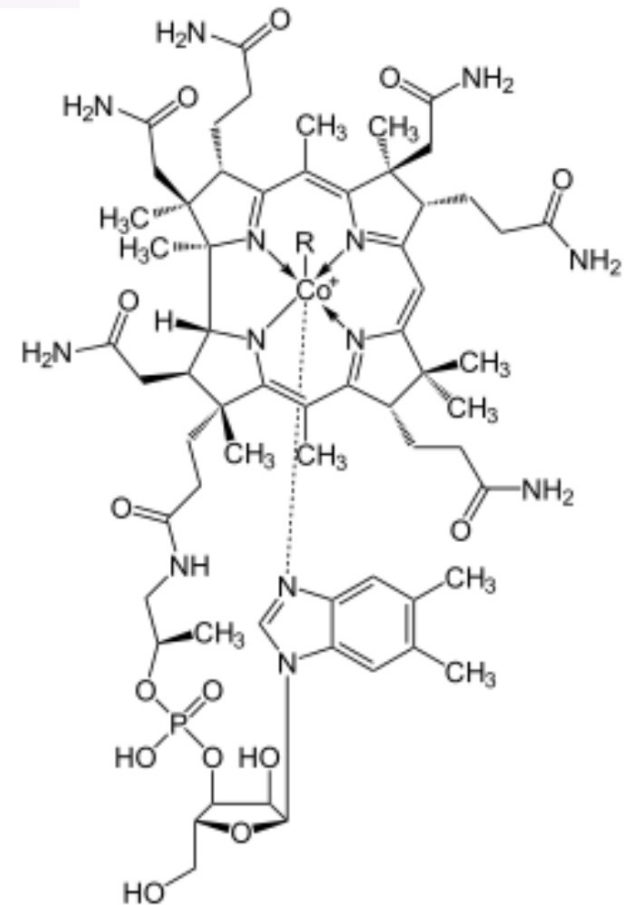
helps maintain
our energy levels

Vitamin B12 contains Cobalt

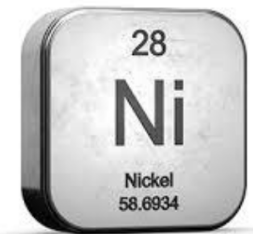


Dorothy Hodgkins
Nobel Prize in Chemistry (1964)

Co-C bond (organometallic)

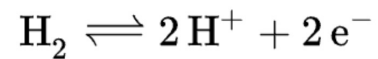


Nickel



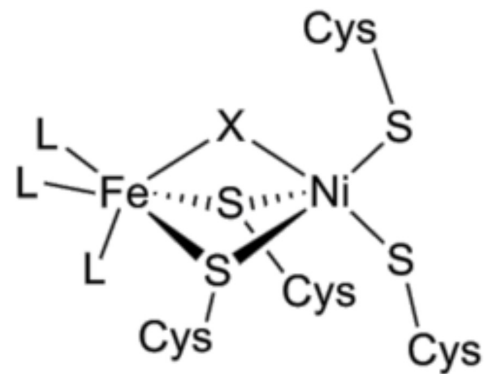
Nickel in the human body is an essential micronutrient and enzyme cofactor, playing roles in RNA structure, glucose utilization, and hormone production

Nickel containing enzymes

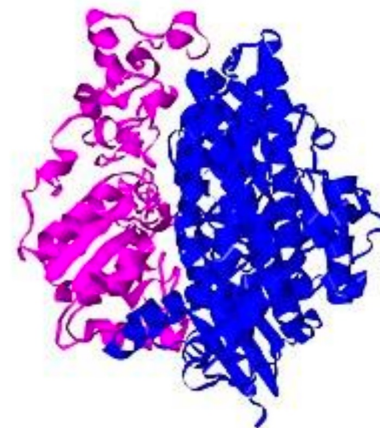


Nickel-Iron hydrogenase

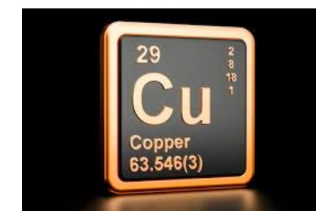
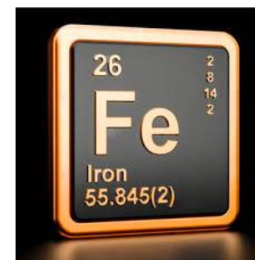
Nickel-Iron-Selenium hydrogenase



L = CO, CN etc



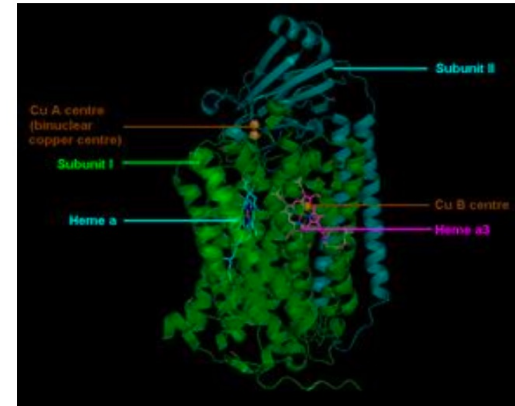
Copper and Iron – the redox metals of the cell



Copper and iron proteins participate in many of the same biological reactions:

- (1) reversible binding of dioxygen, e.g., hemocyanin (Cu), hemerythrin (Fe), and hemoglobin (Fe);
- (2) activation of dioxygen, e.g., dopamine hydroxylase (Cu) (important in the synthesis of the hormone epinephrine), tyrosinases (Cu), and catechol dioxygenases (Fe);
- (3) electron transfer, e.g., plastocyanins (Cu), ferredoxins, and c-type cytochromes (Fe);
- (4) dismutation of superoxide by Cu or Fe as the redox-active metal (superoxide dismutases).

Copper containing metalloproteins



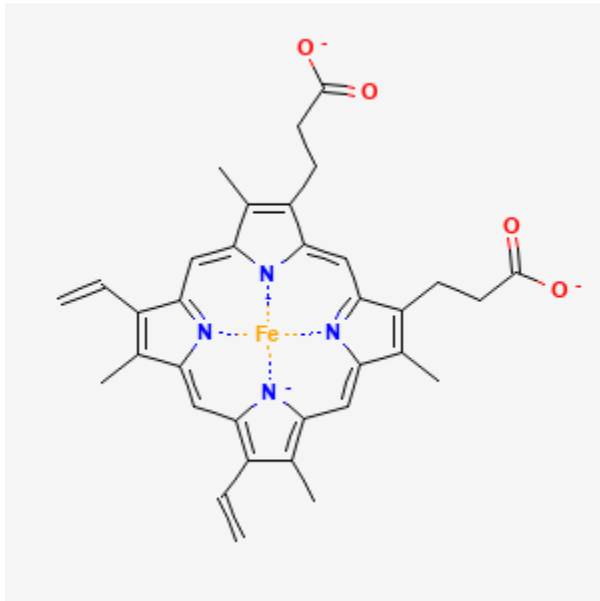
Cu,Zn-superoxide dismutase (SOD1): This enzyme breaks down superoxide anion into hydrogen peroxide and oxygen

Copper amino oxidases (CAOs): These enzymes convert primary amines into aldehydes, producing hydrogen peroxide and ammonia

Cytochrome c oxidase (CCO): This enzyme transports copper from the cytosol to the mitochondria.

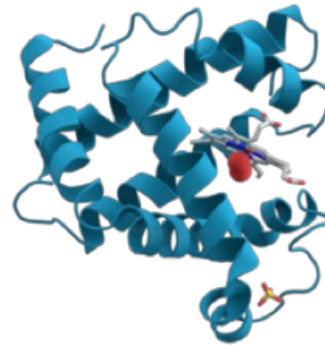
Fe

Myoglobin vs. Hemoglobin Structural Differences



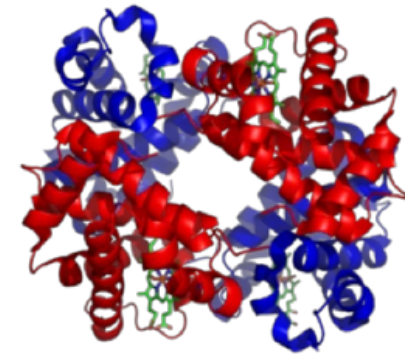
Heme

Myoglobin



- 1 subunit
- 1 heme group

Hemoglobin



- 4 subunits
- 4 heme groups

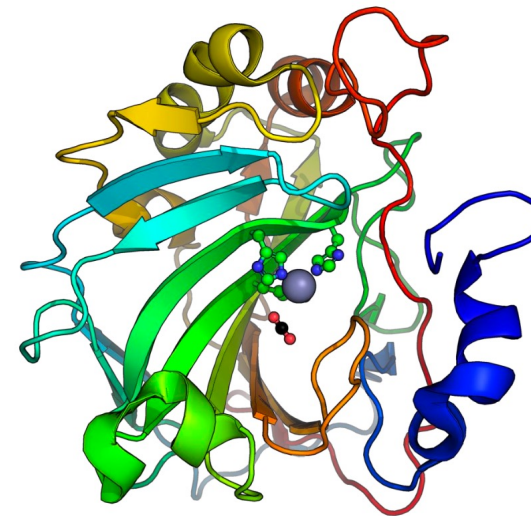
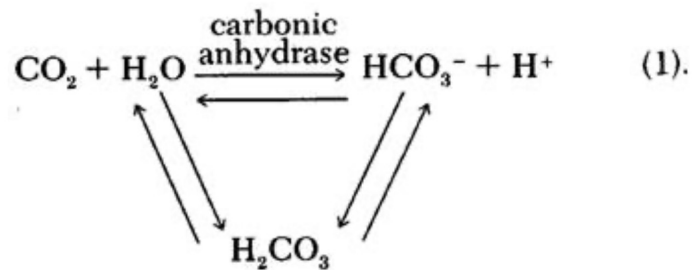
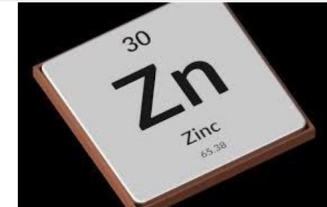
Zinc

Most common transitional metal: Fe

Zn is relatively more abundant (but not a transition metal)

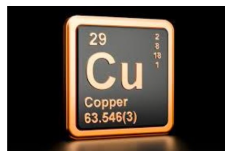
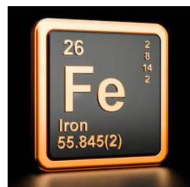
The only trace metal present in all class of enzymes

Structural roles and not catalytic in nucleic acid polymerases and transcription factors



Carbonic Anhydrase

Metal poisoning



In excess, any metal can lead to poisoning